

ANALYSIS BRIEF

Antiwork Topic Modeling

Exploratory longitudinal findings and writing handoff

97,254	199	38.6%	19.0%
filtered posts	non-outlier clusters	topic assignment	theme-map coverage

Bottom line

Across 97,254 management-related r/antiwork posts from March 2021 through February 2025, the updated BERTopic analysis identifies a descriptive shift away from COVID-centered health and safety discussion and toward scheduling and time-off discussion. Compensation, career precarity, and recruitment remain present across the four rolling windows.

What I completed

Rebuilt the analysis with a modern BERTopic pipeline; exported document-level assignments and four rolling-window fits; finalized a 27-cluster, nine-theme codebook; and generated trends, figures, and a reproducibility record.

What the paper should claim

Descriptive prevalence patterns within the documented theme map. The paper should not claim causal effects, full-conversation prevalence, or sentiment change.

Headline patterns

Theme	Y1	Y4	Descriptive readout
Health safety and attendance	4.97%	2.50%	Down 2.47 pp; early COVID/illness concentration contracts.
Scheduling hours and time off	1.71%	3.20%	Up 1.49 pp; largest mapped theme in Y4.
Compensation and pay rights	3.52%	3.04%	Remains substantial in every window after a Y2 high.
Career precarity and exit	2.54%	2.47%	Broadly stable across four rolling windows.

All percentages use every filtered post in each rolling window as the denominator. They describe the mapped subset, not all potentially relevant content in the corpus.

Analysis steps completed

This sequence was completed before the writing handoff. It separates deterministic corpus checks, model-based topic discovery, descriptive prevalence estimates, and robustness evidence.

- 1. Freeze and audit the corpus.** Matched the management-related filter to the preprocessed analysis corpus and confirmed 97,254 unique post IDs. A duplicate audit found 1.2% residual duplicate non-placeholder texts; this is reported rather than silently removed.
- 2. Construct the primary text field.** Used post title plus body after handling deleted and placeholder content. A title-only run was retained as a sensitivity analysis, not substituted for the primary representation.
- 3. Discover topics with a current topic-modeling pipeline.** Embedded documents with all-MiniLM-L6-v2, reduced the embedding space with UMAP, clustered with HDBSCAN, and represented topics with BERTopic c-TF-IDF. The full-corpus model produced 199 non-outlier clusters and assigned 38.6% of posts.
- 4. Document the theme map.** Reviewed the largest 30 clusters and retained 27 interpretable clusters in a nine-theme candidate codebook. Generic, deleted, and community-meta clusters were excluded, yielding 18,506 mapped posts (19.0% of the full filtered corpus).
- 5. Estimate descriptive longitudinal patterns.** Calculated each theme's share of every filtered post in four March-to-February rolling windows. Figure 1 shows the resulting exploratory prevalence patterns; Figure 2 shows monthly assignment coverage so the mapped share is visible.
- 6. Test stability and state the boundary.** Ran seed-stability checks, title-only sensitivity, and unique-post overlap checks across rolling windows. Sentiment was intentionally not analyzed, and the paper will make no causal or full-conversation prevalence claims.

The repository retains the reproducible scripts, codebook, diagnostics, and the evidence used for the brief; the brief itself contains the decisions and figures needed for a writing handoff.

Provisional theme prevalence by rolling window

Share of all filtered posts. Exploratory 27-cluster map excludes generic, deleted, and community-meta clusters.

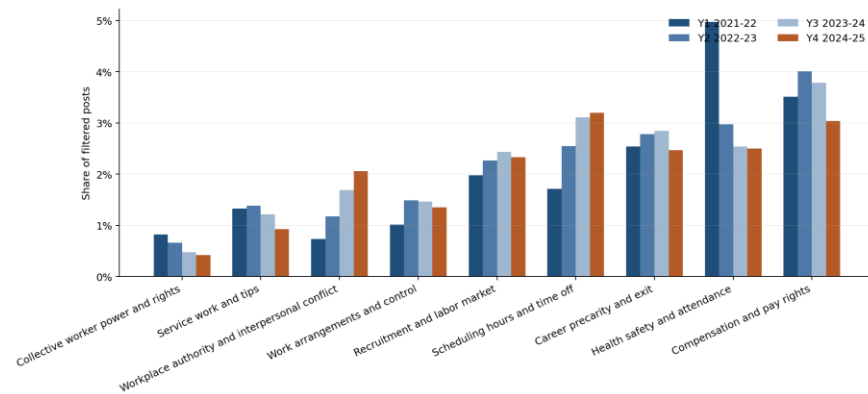


Figure 1. Exploratory prevalence by rolling window. The 27-cluster map excludes generic, deleted, and community-meta clusters.

Method and evidence boundary

Primary method

Sentence-transformer embeddings (all-MiniLM-L6-v2), UMAP, HDBSCAN, BERTopic topic representations, and a full-corpus model used for longitudinal prevalence. Per-window fits are supporting stability diagnostics.

Interpretation boundary

The model assigns 38.6% of posts to non-outlier clusters. The final theme map covers 18,506 posts: 19.0% of all filtered posts and 49.3% of assigned posts.

Coverage is visible, not hidden

Full-corpus BERTopic assignment coverage

97,254 filtered r/antiwork posts, March 2021 through February 2025

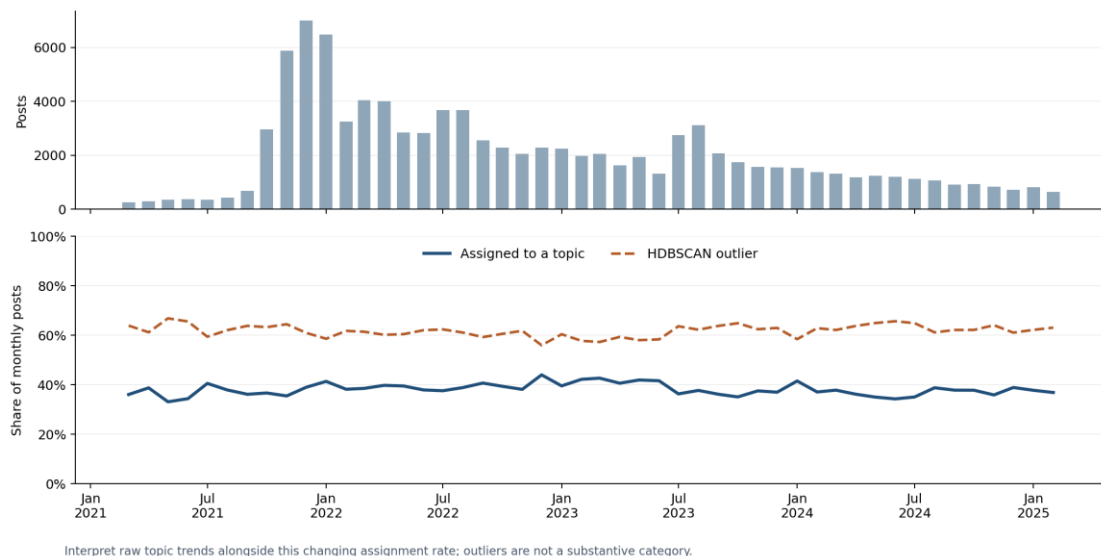


Figure 2. Assignment coverage varied from 33.1% to 44.0% by month. Outliers are a coverage limitation, not a substantive category.

Robustness checks completed

Check	Result
Corpus audit	Inputs contain 97,254 matching unique post IDs; the management-term filter re-matches 100% of retained posts.
Seed stability	Across three UMAP seeds on the same 10,000-post stratified sample, joint-inlier ARI was 0.900 to 0.966.
Window alignment	Conservative one-to-one term matching is supporting context, not proof of substantive persistence.
Text sensitivity	Titles-only clustering was stable but materially different from title-plus-body; title-plus-body remains the primary representation.

A larger all-mpnet-base-v2 encoder check was attempted but did not complete within the available CPU bound, so it is deferred and not treated as evidence.

What is left

Ready now Results, figures, a theme codebook, coverage diagnostics, seed stability, text-mode sensitivity, and a bounded writing frame. The collaborator meeting can focus on interpretation and drafting.	Defer unless claims broaden Sentiment annotation, a larger codebook, saved-model refit, TopicGPT comparison, and the MPNet check are useful extensions, not blockers for this exploratory paper.
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Manuscript status

Draft updated A revised manuscript draft now includes the updated title and abstract; current-study framing and research question; Methods, Results, Discussion, implications, limitations; figures; and primary BERTopic references.	Final editorial pass Before external circulation, harmonize retained Background language with the paper's exploratory, descriptive scope. Historical theory and prior turnover findings should remain context, not claims tested by this study.
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Writing handoff

Draft in this order: method and sampling frame; coverage and descriptive results; limitations; then an exploratory discussion. Use the documented theme map and avoid causal, full-conversation, or sentiment claims.

Reproducible code and shareable evidence: https://github.com/v-tristow_microsoft/antiwork-topic-modeling

Core references: planning/WRITING_HANDOFF.md, planning/ROBUSTNESS.md, and planning/CANDIDATE_THEME_CODEBOOK.csv.

Meeting frame: the updated exploratory analysis is complete and ready to become a bounded, reproducible topic-prevalence paper, without sentiment or causal claims.